Yongzhe (Kindred) Yi

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Education

University of Wisconsin–Madison — Madison, WI
Bachelor of Science in Computer Science — *May 2026*

Publications

[1] **Coordination-Aware Shared Control for Bimanual Dexterous Teleoperation**
Yongzhe (Kindred) Yi, Callie Kim, Mike Hagenow, and Bilge Mutlu — Paper | Code | Video | Poster
(poster) IROS Workshop: Toward Human-Level Robotic Teammates, 2026.
(in submission) IEEE International Conference on Robotics and Automation (ICRA), 2027.

[2] **Measuring and Steering Motion Legibility in Generalist Robot Policies**
Yongzhe (Kindred) Yi, Mike Hagenow, and Bilge Mutlu — Manuscript
(in submission) IEEE Robotics and Automation Letters (RA-L).

[3] **Beyond Task Success: User Interactions with and Perceptions of Hierarchical VLA-Driven Robots**
Xinning He, Callie Kim, **Yongzhe (Kindred) Yi**, Dakota Sullivan, Mike Hagenow, and Bilge Mutlu
(in submission) ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2027.

Research Experience

Undergraduate Researcher / Research Intern — Sep. 2024 – Sep. 2026
People and Robots Laboratory, University of Wisconsin–Madison — *Madison, WI*
NSF HAND Engineering Research Center | *Advisor: Prof. Bilge Mutlu, Mike Hagenow*

- **Vision-Language-Action Policy Learning:** Built an end-to-end $\pi_{0.5}$ fine-tuning and deployment pipeline with **OpenPI**, using the **DROID** setup and **LIBERO** (**MuJoCo**) with **SpaceMouse** for real-robot and simulated data collection, supporting dataset curation and conversion, distributed GPU training, and policy evaluation.
- Developed methods to measure and steer motion legibility in vision-language-action policies, using **classifier-free guidance** (CFG) to generate more legible behaviors while preserving task success. (*See Publications*)
- **Multimodal Teleoperation and Motion Capture:** Experienced with multiple human-motion capture modalities for data collection, including **VR-based tracking** (Meta Quest), **optical motion capture** (OptiTrack), and **RGB/RGB-D vision-based hand tracking** for wrist-pose estimation and dexterous hand retargeting.
- **Bimanual Dexterous Teleoperation and Shared Control:** Built a VR-based teleoperation system for dual Franka Panda arms and Tesollo dexterous hands, implementing custom inverse-kinematics solvers and real-time human-to-robot **motion retargeting**. (*See Publications*)
- Proposed a coordination-aware shared-control framework using an **LSTM** to recognize bimanual coordination patterns and adapt robot assistance; a **16-participant user study** showed reduced operator workload.

Industry Experience

Software Testing Engineer Intern — Jun. 2023 – Sep. 2023
MiiVii Dynamics — *Beijing, China*

- Developed automated testing scripts using Robot Framework and Unix command-line tools to replace repetitive manual testing workflows.
- Conducted functional and serial-port testing to validate stable communication between embedded hardware and software modules.

Research Leadership

Hands and Arms Manipulation Project Lead — Nov. 2025 – Jun. 2026
WiscoHumanoids — *Madison, WI*

- Led a student research team exploring robot-learning methods for dexterous bimanual manipulation of deformable objects, including garment folding tasks based on the LeHome household-manipulation benchmark.
- Developed a dual-arm manipulation platform using a mobile ALOHA system integrated with LeRobot for demonstration collection, policy training, deployment, and evaluation.

Technical Skills

Programming Languages: Python, C++, Java, MATLAB, HTML/CSS
Machine Learning: PyTorch, Diffusion Policy, ACT, Vision-Language-Action (VLA) models, LSTM
Robotics: ROS, ROS2, Motion planning (MoveIt, OMPL), Impedance control, Multi-arm and dexterous teleoperation, Franka Emika Robot, Tesollo Hand, ALOHA, Robotiq, Large-scale datasets (DROID, LIBERO)
XR Integration: VR/AR interfaces for robotics (Meta Quest 3)
Systems and Tools: Linux, Git, Docker, HTCondor, UV, Gazebo, PyBullet, LeRobot, MuJoCo, Open3D